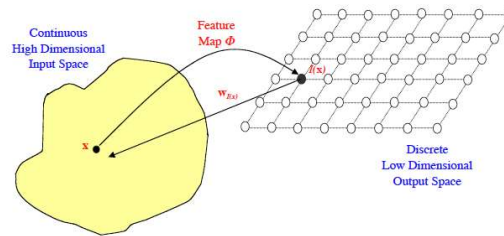


Self Organizing Maps



Statistical Data Mining II
Rachael Hageman Blair

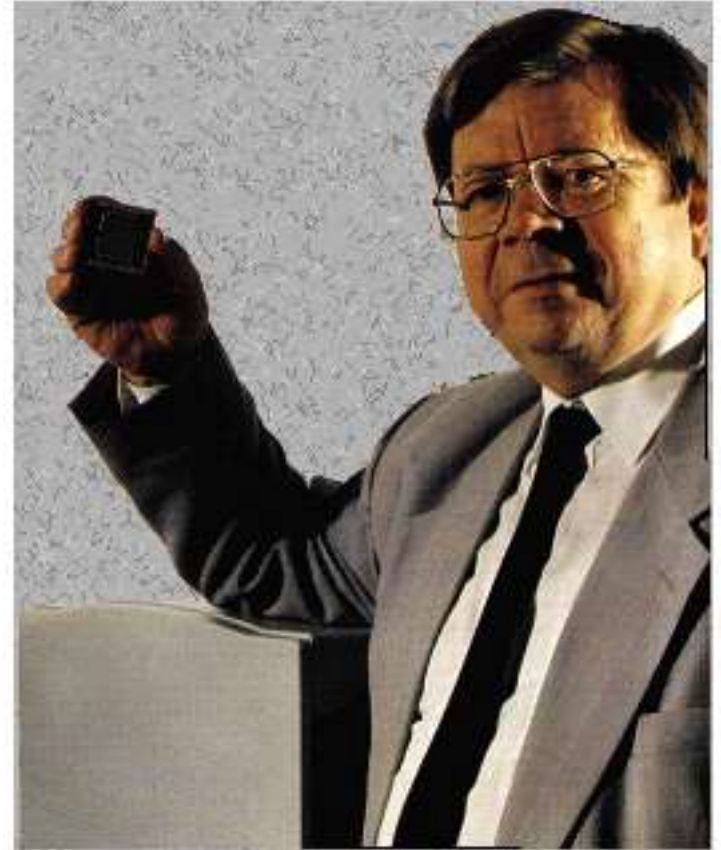
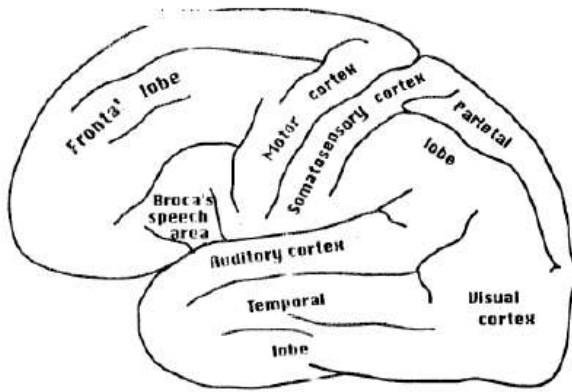
Outline

- Introduction
- Formulation
- Performance
- Visualization
- Relationship to k-means
- Example – Simulation
- Example – Real Data x 2
- Advantages and Disadvantages
- Computation

Teuvo Kohonen

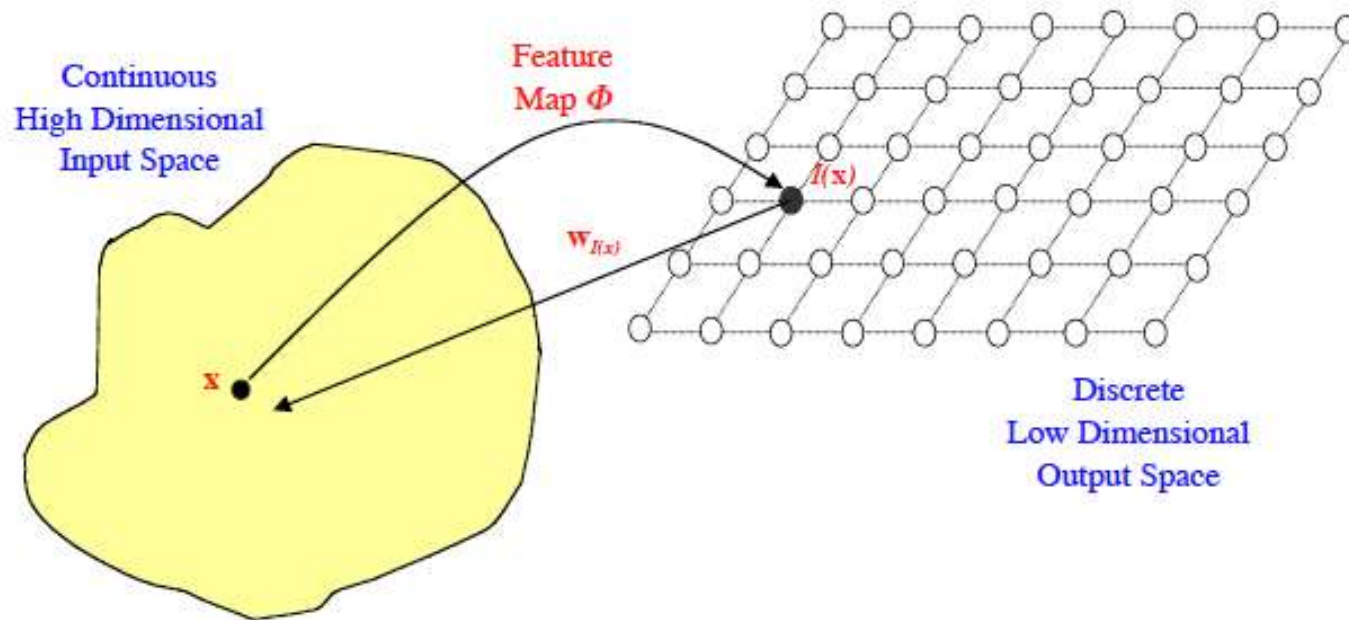
Dr. Eng., Professor of the Academy of Finland
Head, Neural Networks Research Centre,
Helsinki University of Technology, Finland

Maps in Neurobiology/ Neuroscience –
Related neural functions “map” onto identifiable
regions in the brain.



Introduction

- High dimensional data is mapped down to a two dimensional coordinate system.
- Viewed as a constrained k-means where the coordinates are required to lie in a reduced subspace.



Formulation

SOM formulation

- A two-dimensional grid with K prototypes.
- Each of the K prototypes is parameterized with respect to an integer coordinate pair: $l_j \hat{=} Q_1' Q_2$, where $Q_1 = \{1, 2, \dots, q_1\}$ and $Q_2 = \{1, 2, \dots, q_2\}$, and $K = q_1 \times q_2$.
- Denote the K prototypes as $m_j \hat{=} \hat{A}^p$, they are initialized to lie in the two-dimensional principal component plane of the data.
 - Can be thought of as “buttons” that are “sewn” on the principal component plane in a regular pattern.
 - Once the model is fit, the observations are mapped down to a two-dimensional grid.

Formulation

The Mapping

- The observations x_i are processed one at a time.
- Identify the closest prototype m_j to x_i in Euclidean distance in $\hat{A}^p$.
- For all **neighbors** m_k of m_j , move m_k toward x_i through the mapping update:

$$m_k \leftarrow m_k + \overset{\text{Learning rate}}{a}(x_i - m_k).$$

All of the m_j such that the distance between l_j and l_k is small.
Can just use Euclidean distance and “small” is determined by some threshold “ r ” known as the “distance threshold”.

Note: the neighborhood always includes the prototype m_j itself.

Formulation

The Effect of Mapping the prototype space

- We are working in the prototype space $Q_1 \rightarrow Q_2$, the mapping is done accordingly:

$$m_k \rightarrow m_k + a(x_i - m_k).$$

- **What is the effect ?**

We are moving the prototypes closer to the data, while maintaining a smooth two-dimensional spatial relationship between the prototypes.

Performance

Performance depends on a number of factors, including:

- Learning rate a .
- Distance threshold r .
- These parameters are not constant, rather they range over monotonic a spectrum.

At the start, consider broad neighborhoods and map them further away, to making small mappings for tightly knit neighborhoods.

- a is decreased from 1 to 0 incrementally.
- r is decreased from 1 to 0 incrementally.

In practice there are two phases in the “adaptive process”:

- 1. Ordering or self-organizing phase (1000 iterations).**
- 2. Convergence Phase (500 times the number of neurons).**

Performance

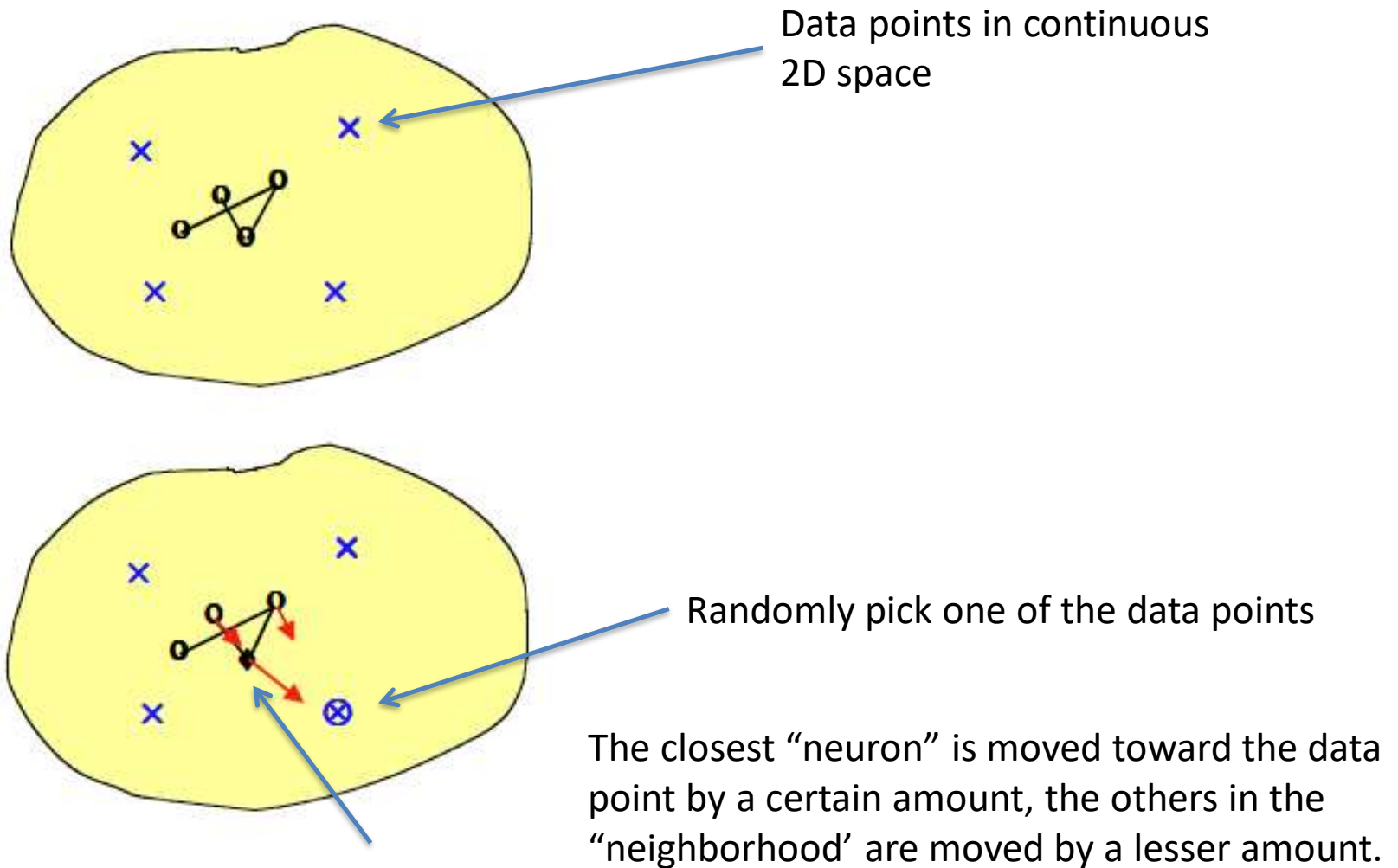
More sophisticated schemes:

$$m_k \leftarrow m_k + ah\left(\|l_j - l_k\|\right)(x_i - m_k).$$

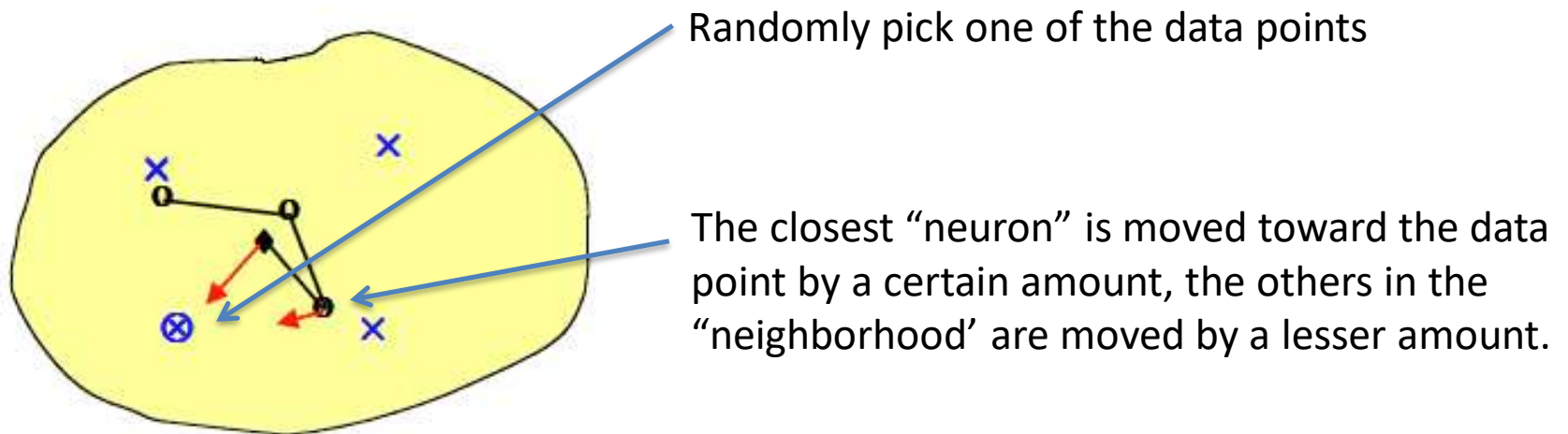
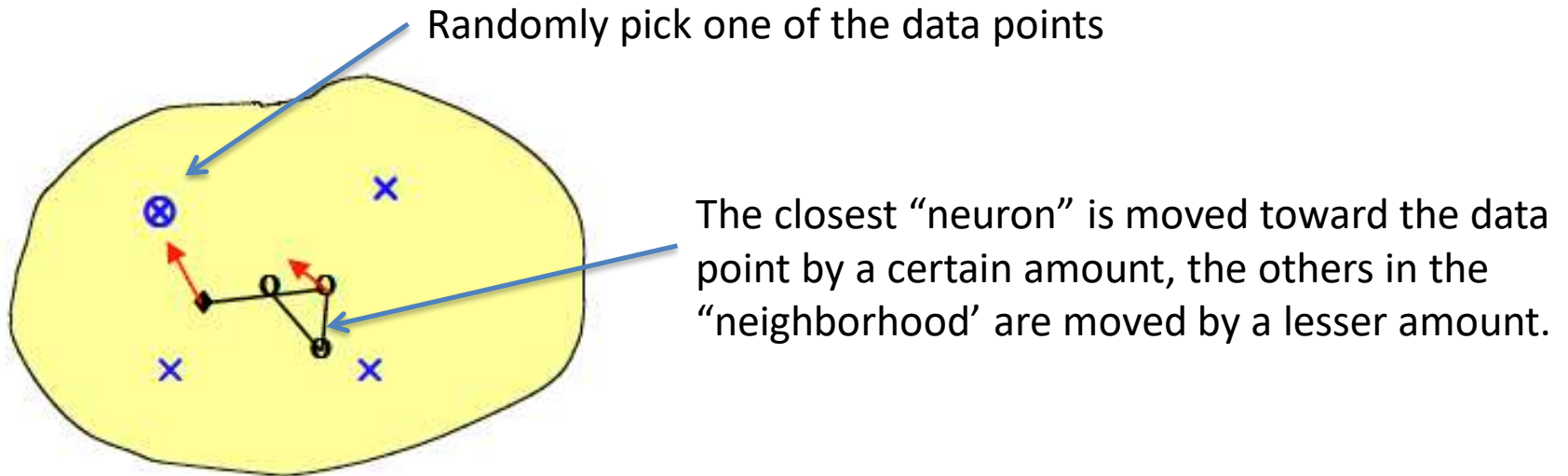

Neighborhood function – gives more weight to prototypes m_k With indices l_k closer to l_j than those further away.

- Would still use r to capture the neighborhood and h for shrinkage.

Visualization

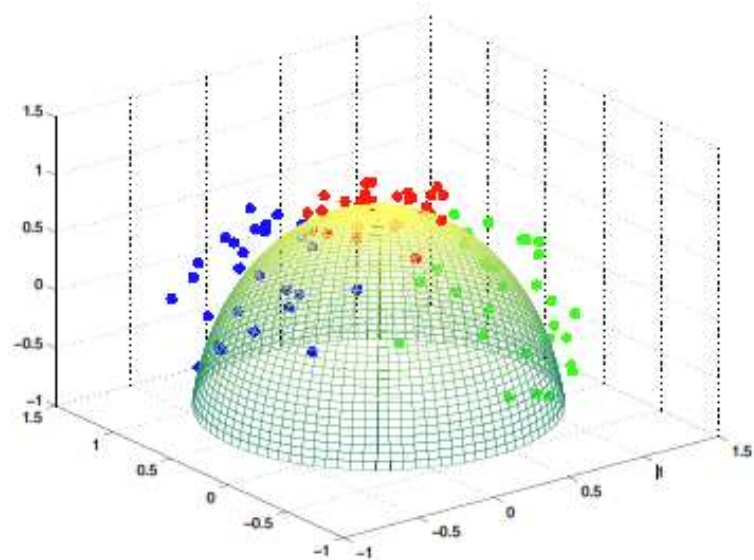


Visualization

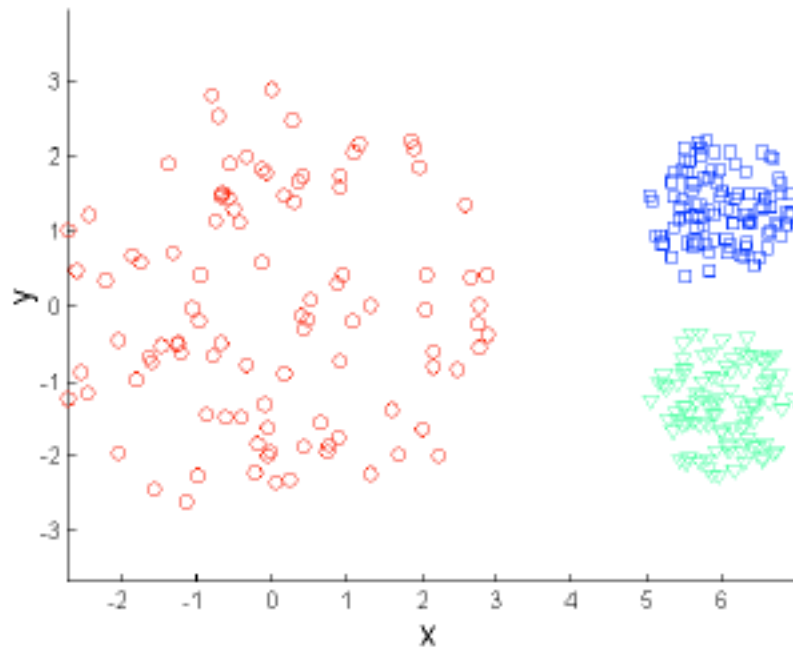


Example - Simulation

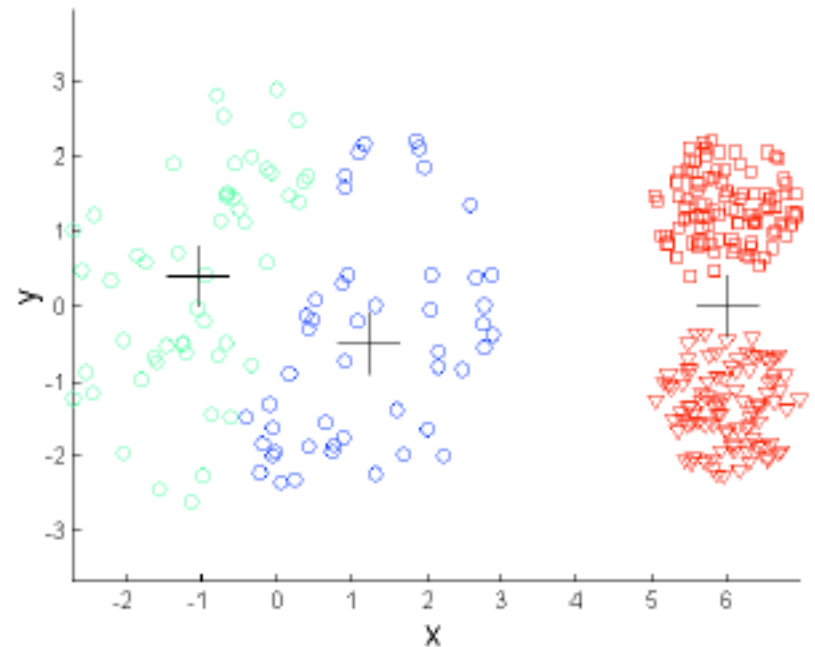
- 90 data points in three dimensions, near a surface of a half sphere of radius 1.
- In three clusters near $(0,1,0)$, $(0,0,1)$ and $(1,0,0)$.
- Red cluster tighter than the others.
- 5x5 grid of prototypes, $R=2$.
- 40 passes through the dataset of 90 observations, with r and α decreasing linearly.



Recall: Kmeans Limitation

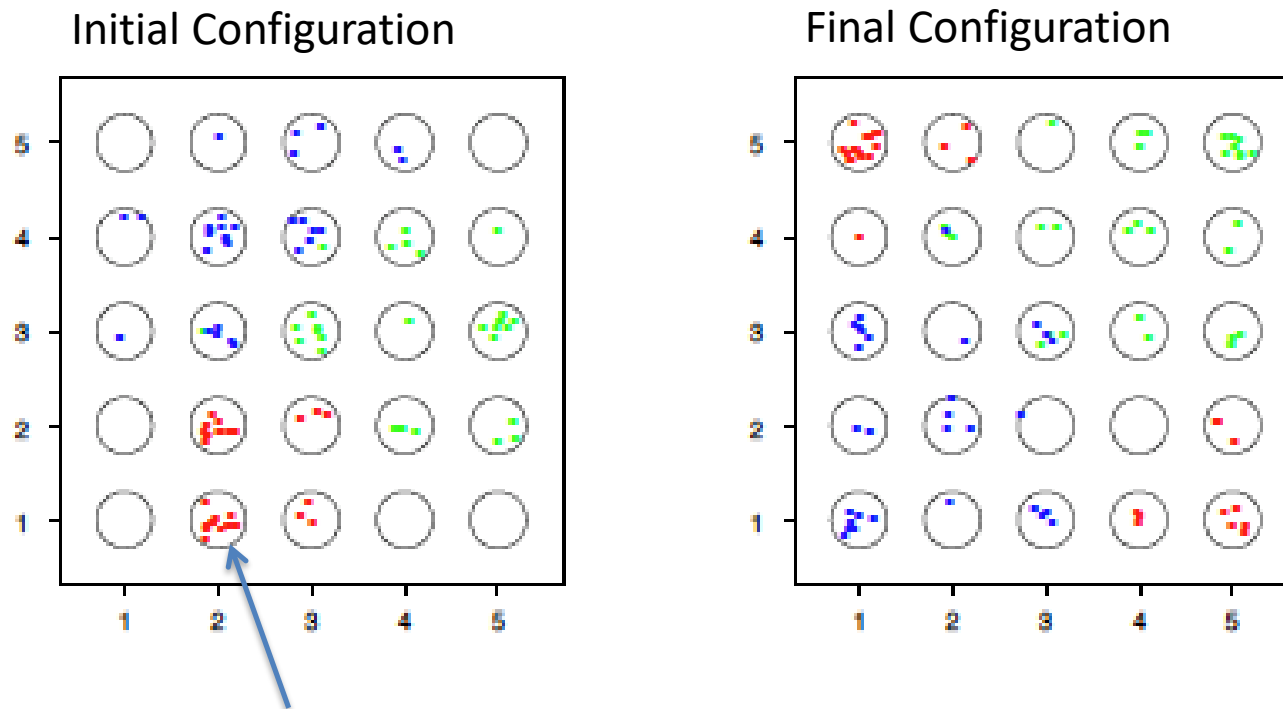


Original Points



K-means (3 Clusters)

Example - Simulation



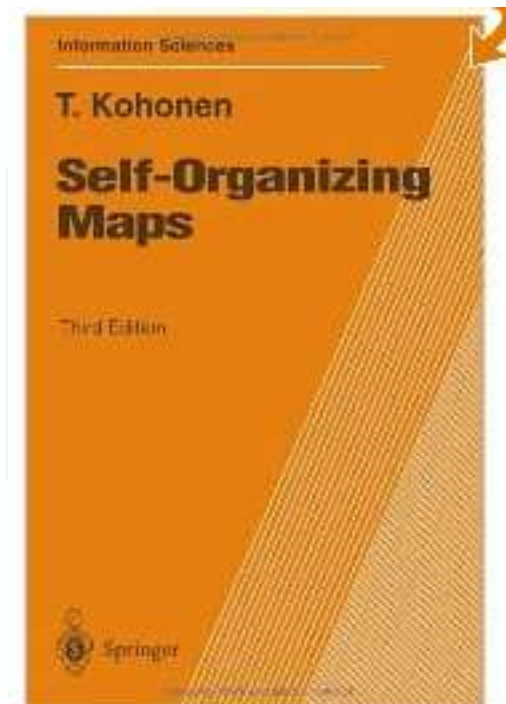
Example – Real Data

- Document retrieval – aims to get the right documents to the right people instantly. Mining of large online databases.
- Finland based effort: SOM fit to document collections.
- Largest collection 7 million patent abstracts. Published in Neural Networks IEEE Transactions.



WEBSOM - Self-Organizing Maps for Internet Exploration

Welcome to test the document exploration tool WEBSOM. An ordered map of the information space is provided: similar documents lie near each other on the map. The order helps in finding related documents once any interesting document is found.



Example – Real Data

- A “term document” matrix, each row represents a document. The entries in each column are the relative frequency of each predefined set of terms.
- These terms could be as large as the set of dictionary entries (50,000 words), or an even larger set of bigrams, or subsets of these.
- Matrices are sparse and can be pre-processed to reduce dimensionality (e.g., SVD).
- Reduced vectors are input into the SOM algorithm.

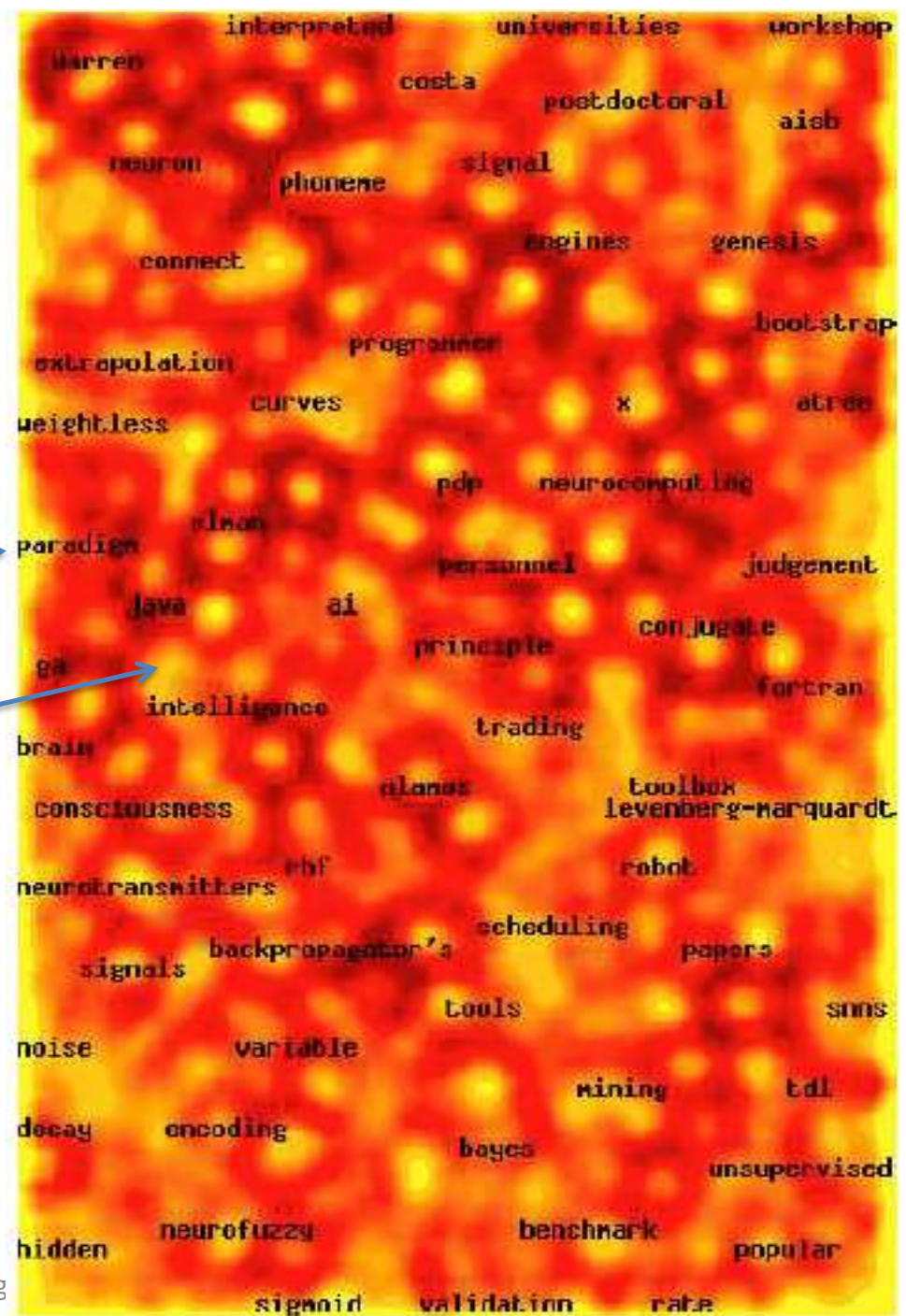
Example – Real Data

Heatmap representation of ~13K
Newsgroup contributions.

Populated nodes
labeled according to
typical content.

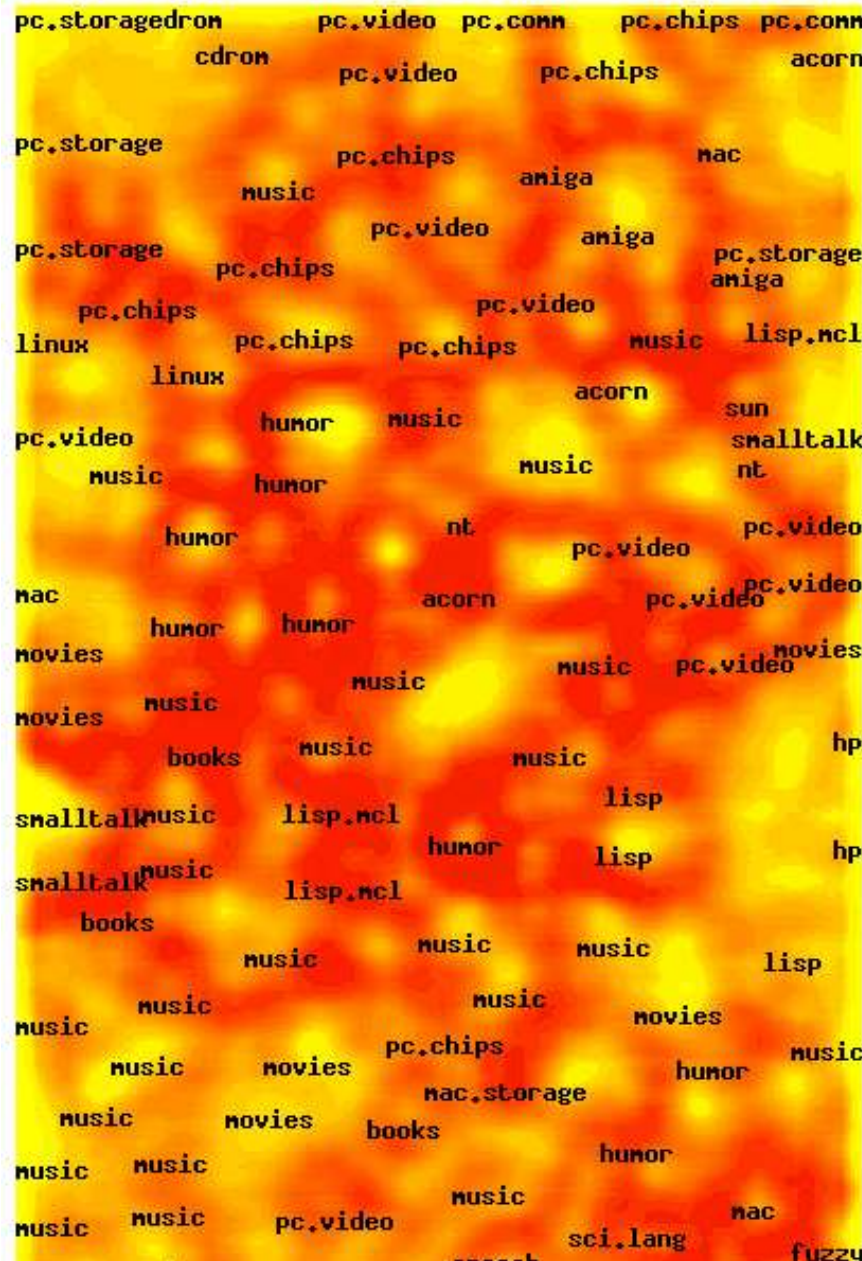
High density regions yellow “bright”

Self Org



WEBSOM map - Million documents

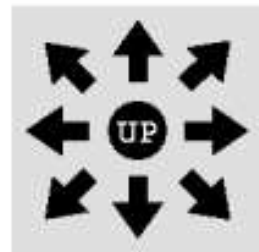
[Instructions](#)



Explanation of the symbols on the map

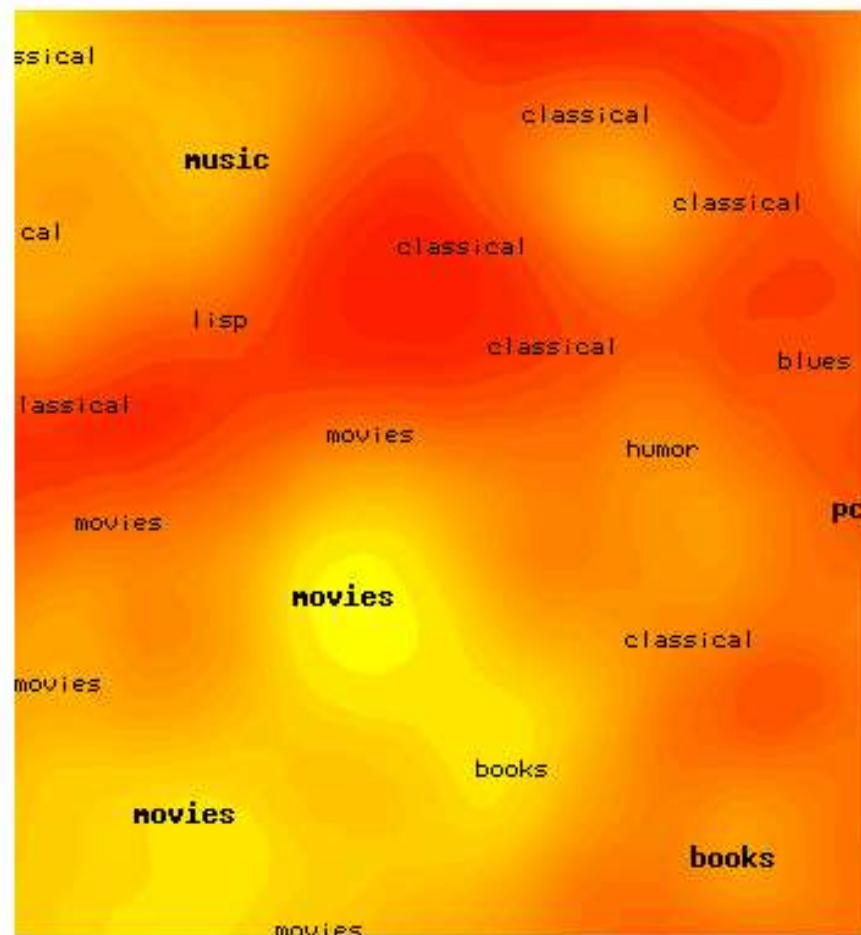
acorn - comp.sys.acorn.hardware
amiga - comp.sys.amiga.hardware
booksEach white20 - rec.arts.books
cdrom - comp.publish.cdrom.hardware
compilers - comp.compilers
fuzzy - comp.ai.fuzzy
genetic - comp.ai.genetic
hp - comp.sys.hp.hardware
humor - rec.humor
lang.eiffel - comp.lang.eiffel
lang.ml - comp.lang.ml
linux - comp.os.linux.hardware
lisp - comp.lang.lisp
lisp.mcl - comp.lang.lisp.mcl
mac - comp.sys.mac.hardware.misc
mac.storage - comp.sys.mac.hardware.storage
movies - movies
music - music
nt - comp.os.ms-windows.nt.setup.hardware
pc.cdrom - comp.sys.ibm.pc.hardware.cd-rom
pc.chips - comp.sys.ibm.pc.hardware.chips
pc.comm - comp.sys.ibm.pc.hardware.comm
pc.storage - comp.sys.ibm.pc.hardware.storage
pc.video - comp.sys.ibm.pc.hardware.video
philosophy - philosophy
plant - bionet.biology.plant
prolog - comp.lang.prolog
sci.lang - sci.lang
smalltalk - comp.lang.smalltalk

WEBSOM zoomed map - Million documents



Click arrows

to move to neighboring areas on the map, and to move up to the overall view.



blues - rec.music.bluenote

books - rec.arts.books

classical - rec.music.classical

humor - rec.humor

lisp - comp.lang.lisp

movies - rec.arts.movies.past-films

music - music

rec.arts.books

[Re: Top 5 Things Which Make Me Suspicious About the Book](#) ♦ Rich Gibson, Tue, 22 Aug 1995, Lines: 25.
[Re: Suggestions for "nature" novels wanted](#) ♦ Twitch, Thu, 14 Dec 1995, Lines: 16.
[Re: Humorous novels?](#) ♦ Karen. M. Girard Weiss, Thu, 27 Jun 1996, Lines: 25.
[Re: Help with Hermann Hesse .](#) ♦ Harish P Cherukuri, 11 Jul 1996, Lines: 19.
[Re: Indian Books](#) ♦ Ms Tory Caplan, 25 Jul 1996, Lines: 30.
[Re: suggest "coming of age" novels?](#) ♦ Kirk A. Weiss, Tue, 06 Aug 1996, Lines: 16.
[Re: Bought some skinny books today](#) ♦ Matthew P Wiener, 29 Aug 1996, Lines: 12.

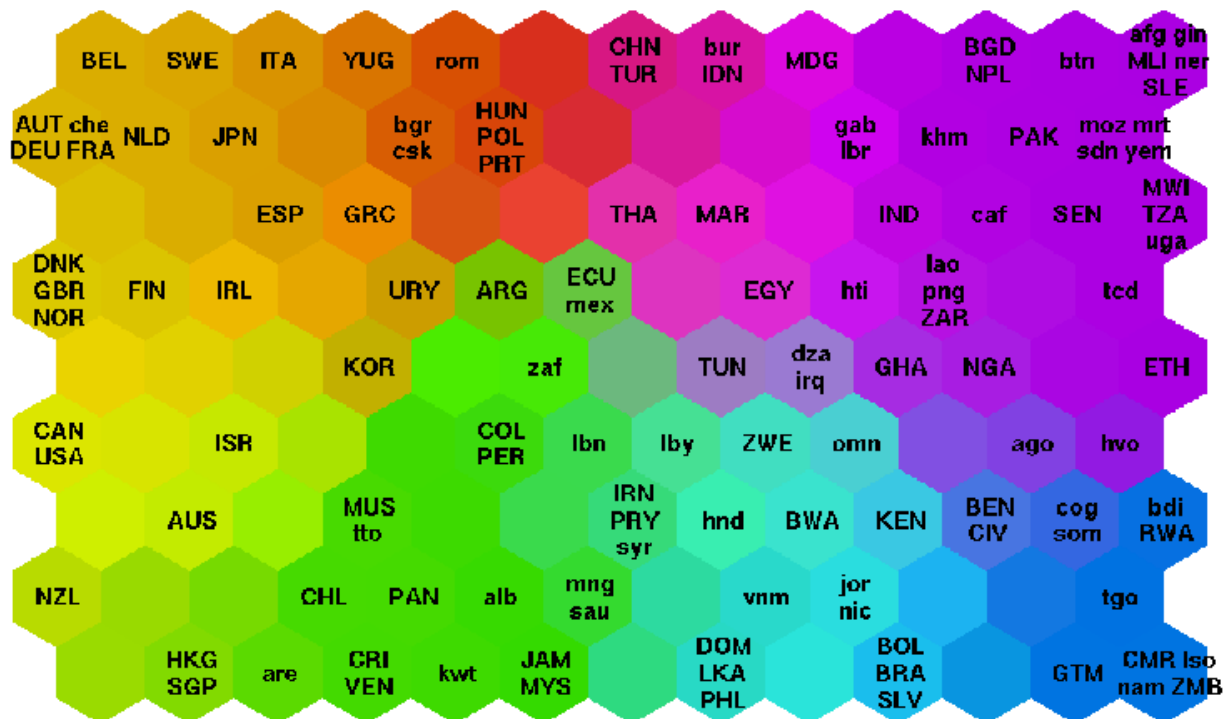
rec.arts.movies.past-films

[Re: Help: Obscure 1960s Japanese fantasy film?](#) ♦ Madhu Krishnaswamy, 26 Aug 1995, Lines: 4.
[Re: Godard](#) ♦ lang thompson, 5 Sep 1995, Lines: 40.
[Re: INGMAR BERGMAN's Fanny & Alexander QUESTION](#) ♦ Keith Howie, Tue, 05 Sep 1995, Lines: 28.
[New Zealand films](#) ♦ Ellen E Aspengren, 4 Oct 95, Lines: 5.
[Re: Blackhawk films address & Hal Roach shorts](#) ♦ Ron Maher, 14 Jan 1996, Lines: 40.
[Re: Please Help: Films with Tracking Shots](#) ♦ Keith Howie, Thu, 01 Feb 1996, Lines: 15.
[Book on Stanwyck films](#) ♦ MJS at EOE, 9 Feb 1996, Lines: 5.
[Re: What's the big deal about CITIZEN KANE?](#) ♦ FilmGene, 29 Feb 1996, Lines: 13.
[Re: Barbara hershey Films](#) ♦ PJK, Mon, 22 Apr 1996, Lines: 52.
[Re: tarantino is pulling an Orson Welles](#) ♦ Brian Varney, Wed, 19 Jun 1996, Lines: 14.
[Diane Kurys/Love After Love](#) ♦ chelsea corazon, 12 Jul 1996, Lines: 22.
[Re: FILMS WHICH ARE ALLEGORIES](#) ♦ Walter Ulrich, 4 Aug 1996, Lines: 16.

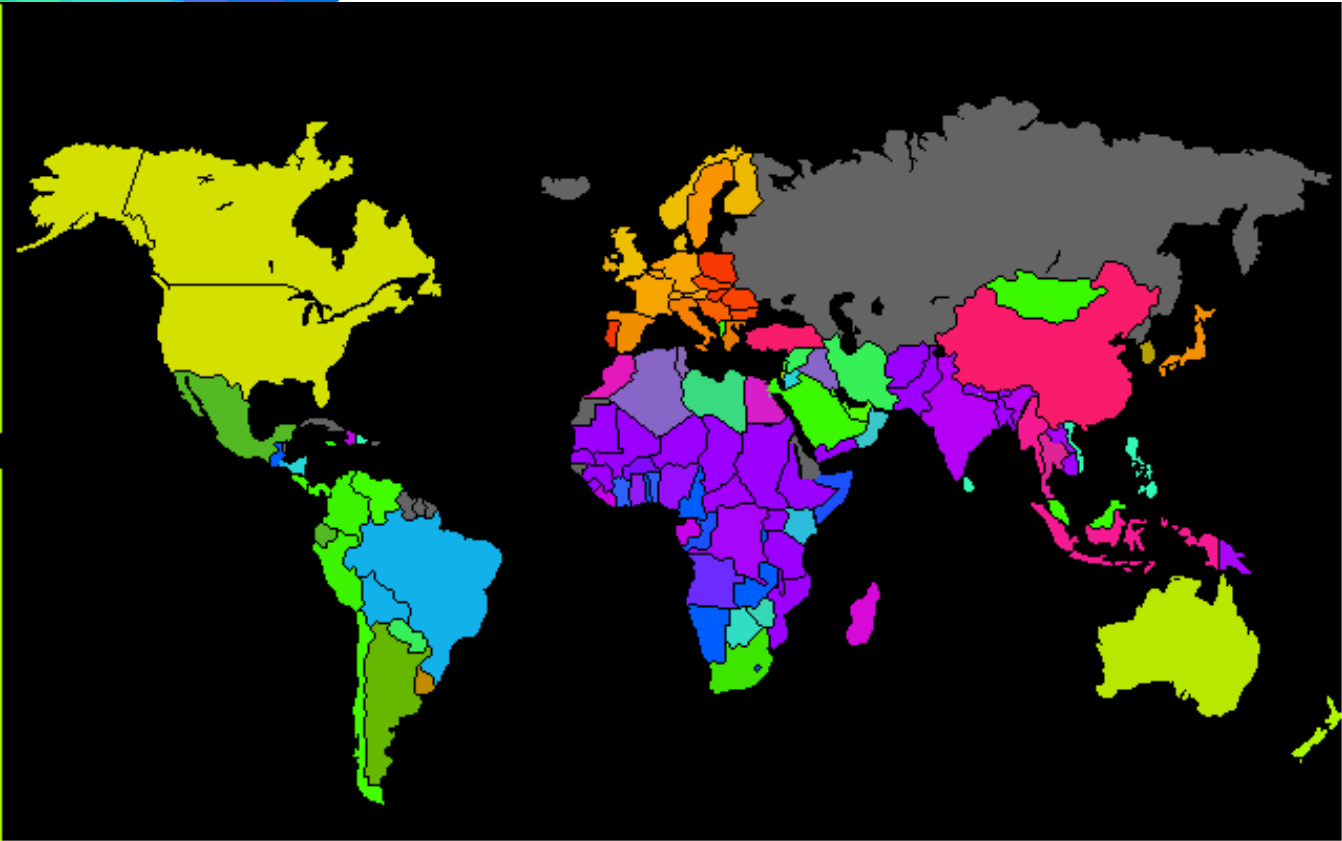
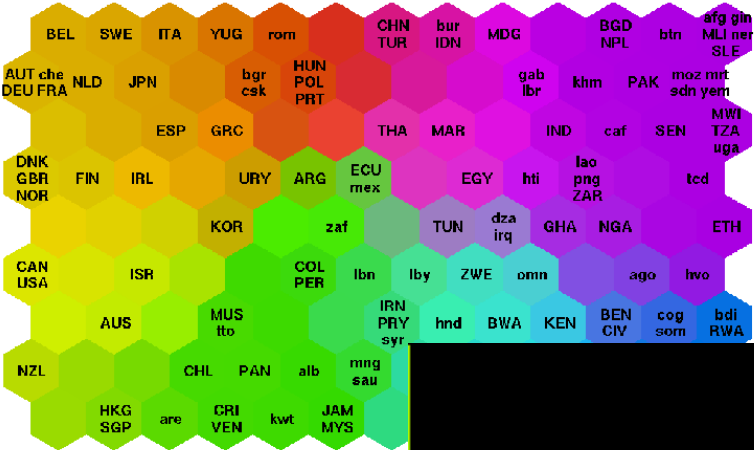
Example – Poverty Map

- Data – world bank statistics of countries in 1992.
- 39 indicators of quality of life factors such as state of health, nutrition, educational services etc.

Countries organized on a self-organizing map based on indicators related to poverty:



Countries organized on a self-organizing map based on indicators related to poverty:



Self Organizing Maps

Advantages

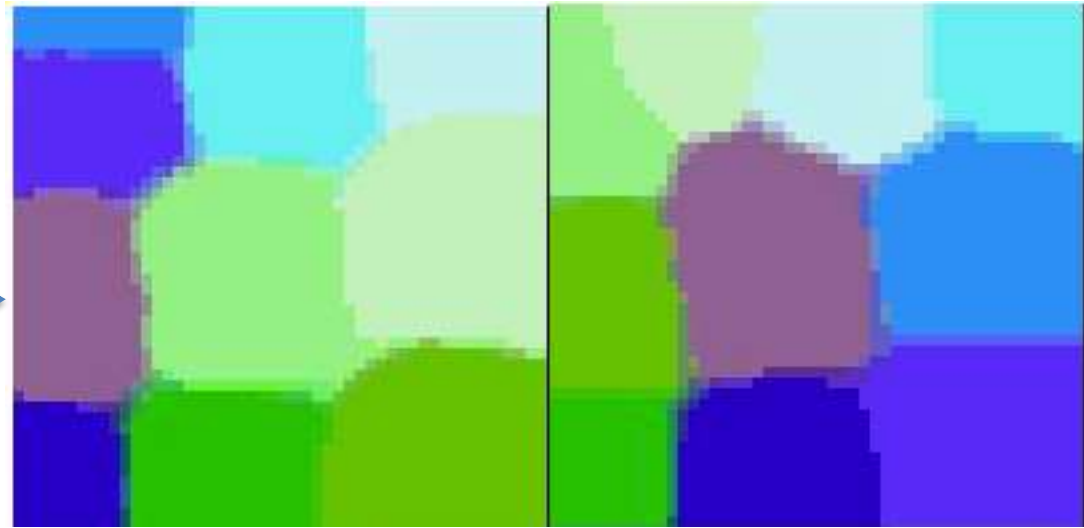
- Easily interpreted. Grid clustering makes it easy to understand similarities in the data.
- Can be used in a supervised or unsupervised context. E.g., supervised – map emails onto a grid, clusters represent spam and not spam.
- Recent advances make large-scale problems manageable, but not easy yet in terms of software.

Drawbacks

- Not able to handle missing data.
- Computationally expensive.
- Lack of data or extraneous data will add randomness with the groupings.
- Finding the correct data requires determination of which factors are “relevant” for the groupings. Difficult or impossible for many situations.
- Clusters can easily be divided into smaller clusters and put in different areas. May be avoided by good initialization.

Drawbacks

Three shades
of purple. Two
Dark split by
One light.



Should be next to one another

Using colors – we can detect this by eye. In real data, it will be impossible to detect, are they related are they not, who knows. If we knew the groupings we would not be clustering in the first place.

Computation

kohonen: Supervised and unsupervised self-organising maps

Supervised and unsupervised self-organising maps

Version: 2.0.10
Depends: R ($\geq 2.6.0$), [class](#), [MASS](#)
Published: 2012-08-13
Author: Ron Wehrens
Maintainer: Ron Wehrens <ron.wehrens at
License: [GPL \(\$\geq 2\$ \)](#)
Citation: [kohonen citation info](#)
In views: [ChemPhys](#), [Multivariate](#)
CRAN checks: [kohonen results](#)

Downloads:

Package source: [kohonen 2.0.10.tar.gz](#)
MacOS X binary: [kohonen 2.0.10.tgz](#)
Windows binary: [kohonen 2.0.10.zip](#)



Journal of Statistical Software

October 2007, Volume 21, Issue 5.

<http://www.jstatsoft.org/>

Self- and Super-organizing Maps in R: The kohonen Package

Ron Wehrens
Radboud University Nijmegen

Lutgarde M. C. Buydens
Radboud University Nijmegen

Search R2012b Documentation

Neural Network Toolbox Clustering

selforgmap

Self-organizing map

Syntax

```
selforgmap(dimensions, coverStep
```

Description

Self-organizing maps learn to cluster data based on similarity, topology, with a preference (but no

Abstract

In this age of ever-increasing data set sizes, especially in the natural sciences, visualisation becomes more and more important. Self-organizing maps have many features that make them attractive in this respect: they do not rely on distributional assumptions, can handle huge data sets with ease, and have shown their worth in a large number of applications. In this paper, we highlight the **kohonen** package for R, which implements self-organizing maps as well as some extensions for supervised pattern recognition and data fusion.

Keywords: self-organizing maps, visualisation, classification, clustering.